第十三讲 弯曲变形

作业：7-1(d),3,4(b)，11，12，14(a)，16；

第七章　弯曲变形

7.1 挠度与转角

7.2 挠曲轴近似微分方程

7.3 计算梁位移的积分法

7.4 计算梁位移的叠加法

7.5 简单静不定梁

7.6 梁的刚度条件与合理刚度设计

第七章 弯曲变形

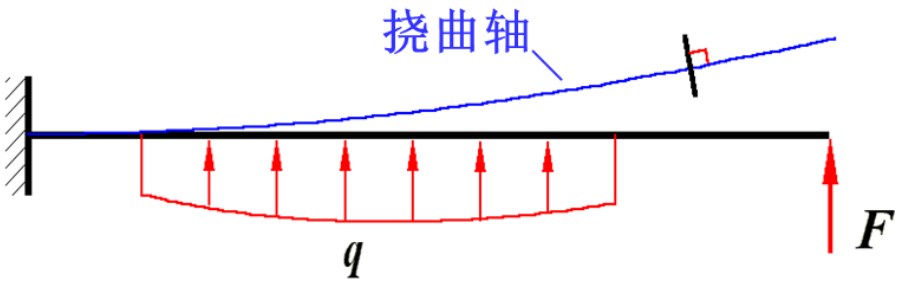
7.1 挠度与转角

1、弯曲变形的特点

2、挠度与转角

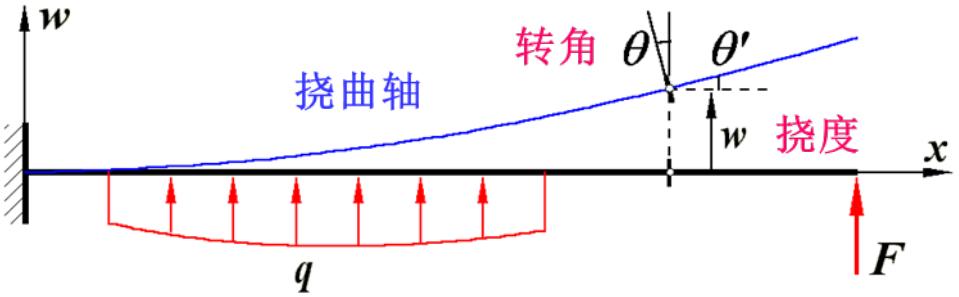
3、认识微段变形

1、弯曲变形的特点



- 直梁，轴线变为曲线。
- 变弯后的轴线，称为挠曲轴。
- 挠曲轴：连续、光滑的曲线。
- 对称弯曲，挠曲轴位于纵向对称面内。
- 对于细长梁，剪力的影响很小，弯曲平面假设成立，横截面保持为平面、与挠曲轴正交

2、挠度与转角



挠度

横截面形心在垂直于梁轴方向上的位移；

$w = w(x)$ —挠曲轴方程

转角

转角：横截面的转角

$\theta = \theta(x)$ —转角方程

挠度与转角

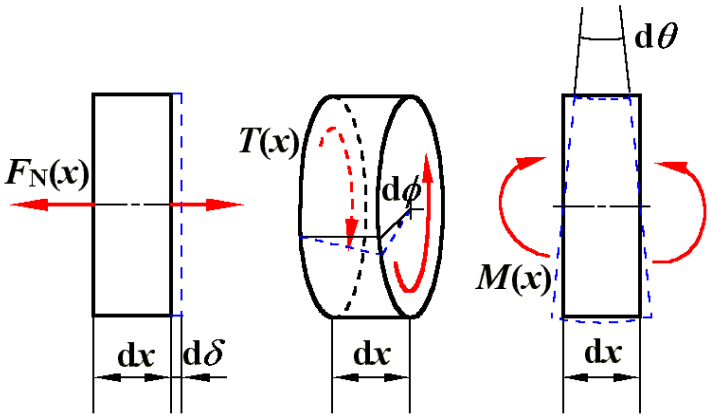
1）小变形、不计剪切：

横截面与挠曲轴正交，

$$\theta(x) = \frac{\mathrm{d}w(x)}{\mathrm{d}x}$$

2）计及剪切：挠度与转角独立。

3、认识微段变形



三个平面假设

微段变形

应力

$$d\delta = \frac{F_N}{EA}dx,\quad d\varphi = \frac{T}{GI_P}dx$$

$$d\theta = \frac{dx}{\rho} = \frac{M(x)}{EI_z}dx$$

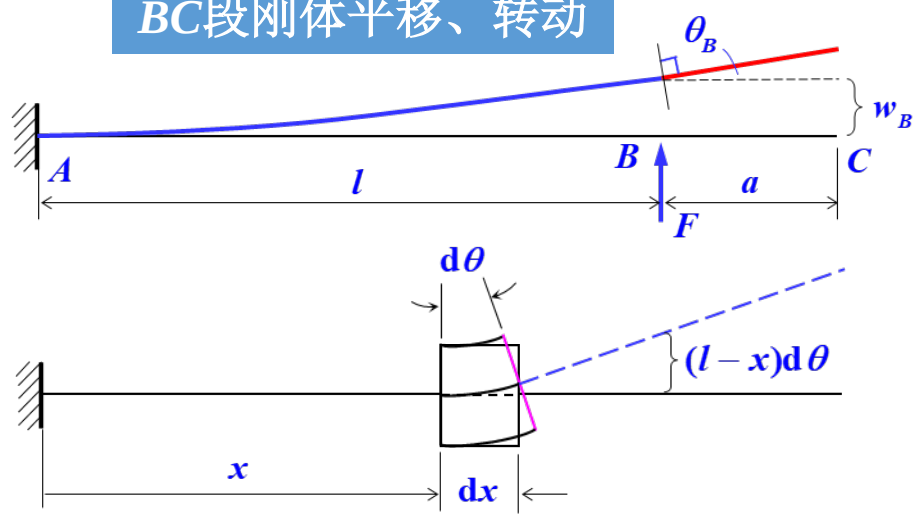
$$\Delta l = \int d\delta$$

$$\Delta\varphi = \int d\varphi$$

梁变形？

揭示了拉压弯扭的基本特征

BC段刚体平移、转动



$$d\theta_B = d\theta,\quad dw_B = (l-x)d\theta$$

影响系数——微段变形的贡献

$$\mu_{\theta B} = 1,\quad \mu_{wB} = l - x$$

$$\theta_B = \int_0^l \mu_{\theta B}\,d\theta,\quad w_B = \int_0^l \mu_{wB}\,d\theta$$

$$\theta_C = \theta_B,\quad w_C = w_B + \theta_B a$$

微段变形累加形成截面挠度与转角

第七章 弯曲变形

7.2 挠曲轴近似微分方程

1、挠曲轴微分方程

2、挠曲轴近似微分方程

3、位移边界条件与连续光滑条件

4、直观变形分析

1、挠曲轴微分方程

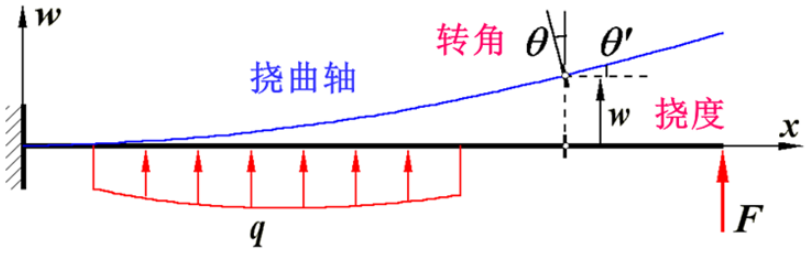
对称弯曲梁中性层曲率

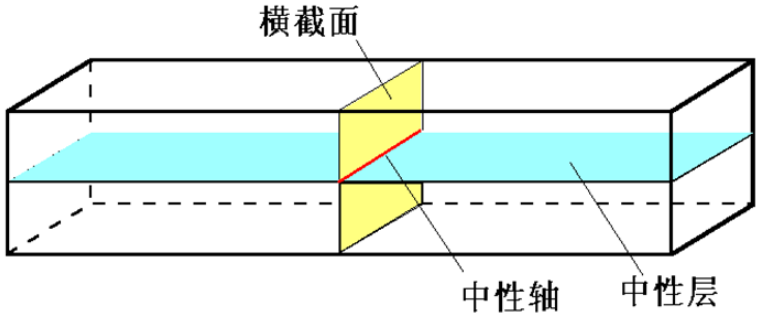
纯弯：
$$\frac{1}{\rho}=\frac{M}{EI}$$

非纯弯细长梁，不计剪切：

$$\frac{1}{\rho(x)}=\frac{M(x)}{EI}$$

——挠曲轴$w(x)$的曲率





平面曲线$w=w(x)$的曲率

$$\frac{1}{\rho(x)}=\pm\frac{w''}{\left[1+w'^2\right]^{3/2}}$$

挠曲轴微分方程

$$\frac{w''}{\left[1+w'^2\right]^{3/2}}=\pm\frac{M(x)}{EI}$$

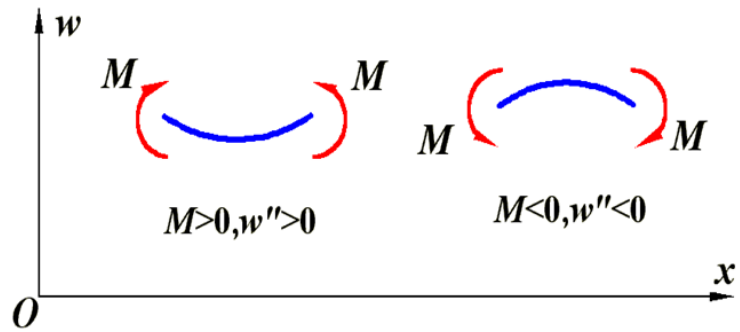
- w—弯矩引起的挠度
- $\sigma_{max}\le\sigma_p$

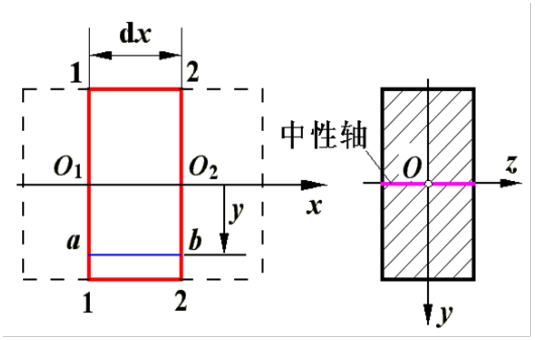
2、挠曲轴近似微分方程

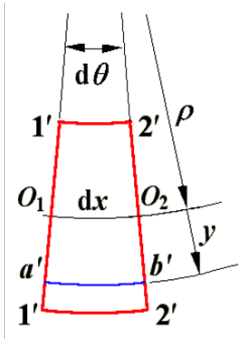
小变形情况， $w'^2 << 1$

$$\frac{w''}{\left[1+w'^2\right]^{3/2}} = \pm\frac{M(x)}{EI}$$

$$\Rightarrow \quad \frac{d^2w}{dx^2} = \frac{M(x)}{EI}$$







- 小变形，线弹性（$\sigma_{max} \le \sigma_p$）
- 挠度向上为正，弯矩上弯为正。

从微段变形看

小变形
不计剪切

$$d\theta = \frac{dx}{\rho} = \frac{M(x)}{EI}dx \quad \Rightarrow \quad \frac{d\theta}{dx} = \frac{d}{dx}\left(\frac{dw}{dx}\right) = \frac{d^2w}{dx^2} = \frac{M(x)}{EI}$$

2、挠曲轴近似微分方程

挠度转角与挠曲轴

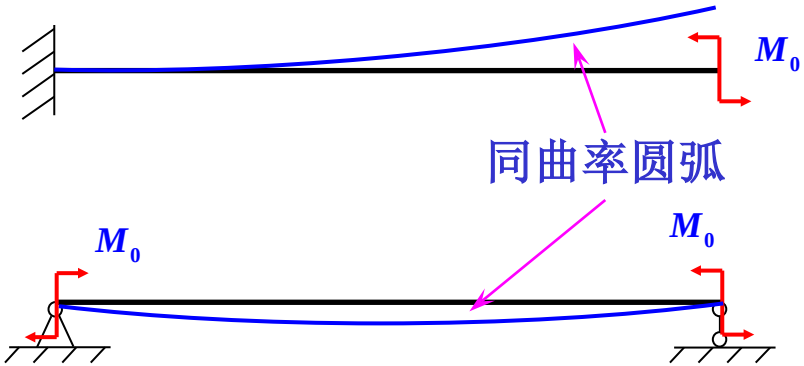
弯矩 → 微段转角 挠曲轴曲率 → 挠曲轴 形状

支座 → 边界条件 连续条件 → 挠曲轴 位置

→ 梁截面 挠度转角

$$\frac{d^2w}{dx^2}=\frac{M(x)}{EI}$$

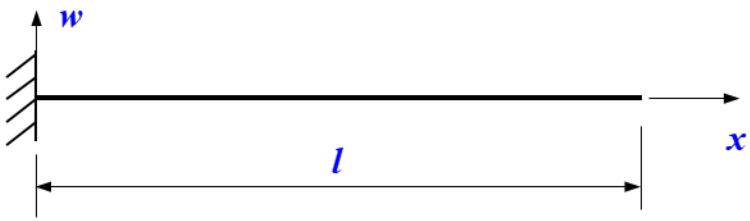
弯矩定形状，支座定位置



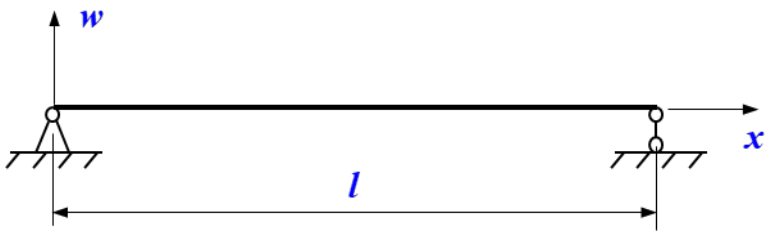
挠曲轴形状相同，位置不同

3、位移边界条件与连续光滑条件

- 位移边界条件：支座（约束）处位移应满足的条件；
- 连续光滑条件：梁段交接处位移应满足的条件。

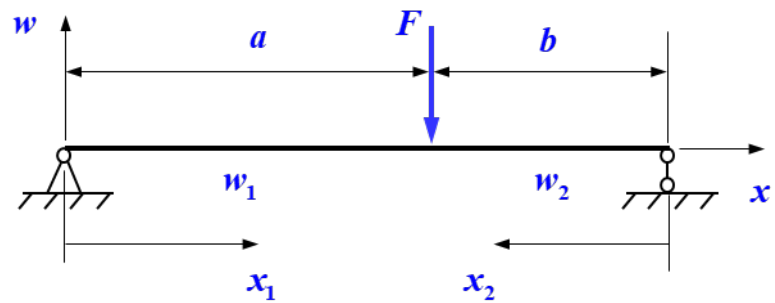


$$x=0,\quad w=0,\quad \frac{\mathrm{d}w}{\mathrm{d}x}=0$$



$$x=0,\quad w=0;$$

$$x=l,\quad w=0$$



$$x_1=0,\qquad w_1=0;$$

$$x_2=0,\qquad w_2=0;$$

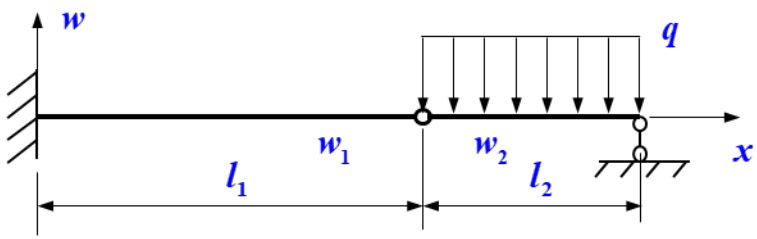
$$x_1=a,\ x_2=b,\qquad w_1=w_2;$$

$$\frac{\mathrm{d}w_1}{\mathrm{d}x_1}=-\frac{\mathrm{d}w_2}{\mathrm{d}x_2}$$

光滑条件即为转角连续

3、位移边界条件与连续光滑条件

- 位移边界条件：支座（约束）处位移应满足的条件；
- 连续光滑条件：梁段交接处位移应满足的条件。



$$x = 0, \qquad w_1 = 0;$$

$$\frac{\mathrm{d}\,w_1}{\mathrm{d}\,x} = 0;$$

$$x = l_1 + l_2, \qquad w_2 = 0;$$

$$x = l_1, \qquad w_1 = w_2.$$

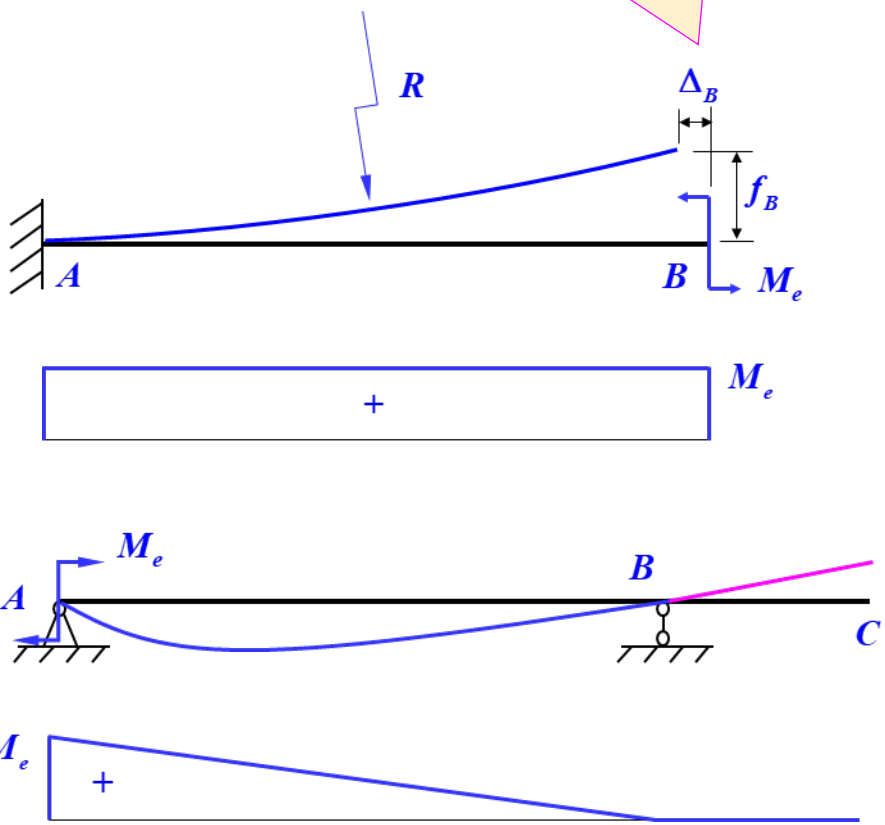
4、直观变形分析

绘制挠曲轴大致形状

- 直观判断梁变形的基本特征；
- 挠曲轴凸凹、挠度和转角的方向等。

弯矩定形状
支座定位置

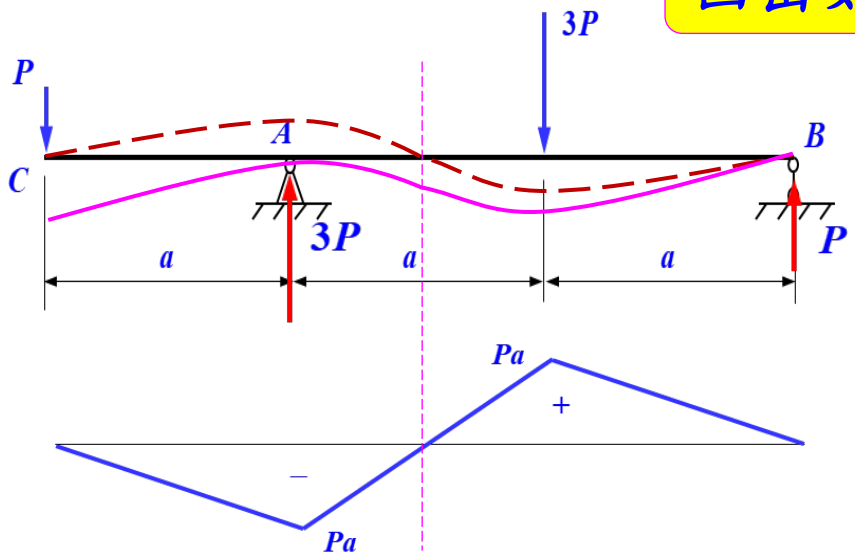
挠曲轴（圆弧）长度不变
轴向位移有多大？

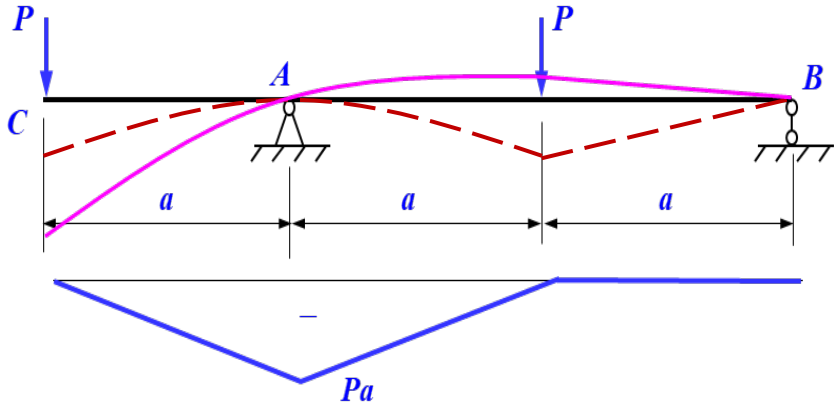


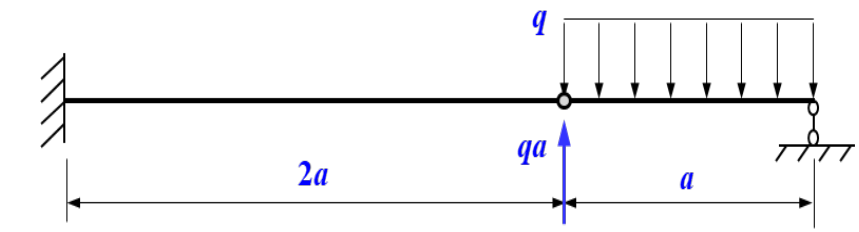
4、直观变形分析

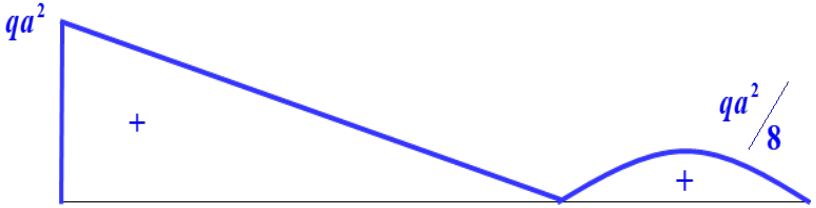
弯矩定形状
支座定位置

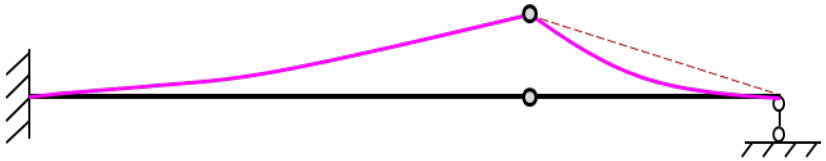
画出如下梁的大致形状











第七章 弯曲变形

7.3 计算梁位移的积分法

1、挠曲轴近似微分方程的积分

2、例题

1、挠曲轴近似微分方程的积分

$$\frac{\mathrm{d}^2 w}{\mathrm{d}x^2}=\frac{M(x)}{EI}$$

$$\theta(x)=\frac{\mathrm{d}w}{\mathrm{d}x}=\int\frac{M(x)}{EI}\mathrm{d}x+C$$

$$w(x)=\iint\frac{M(x)}{EI}\mathrm{d}x+Cx+D$$

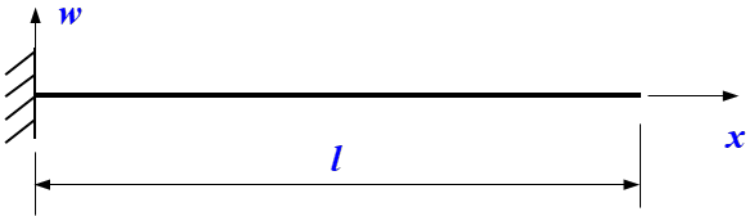
挠曲轴形状

刚体转动

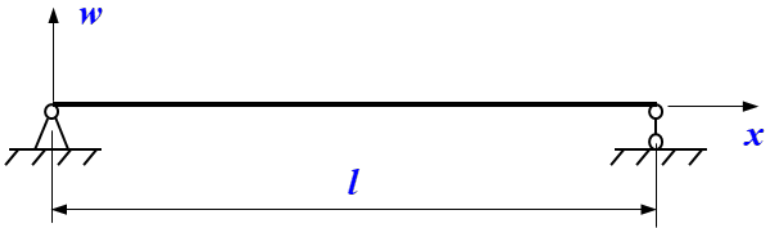
刚体平移

弯矩

支座



$$x=0,\quad w=0,\quad \frac{\mathrm{d}w}{\mathrm{d}x}=0$$

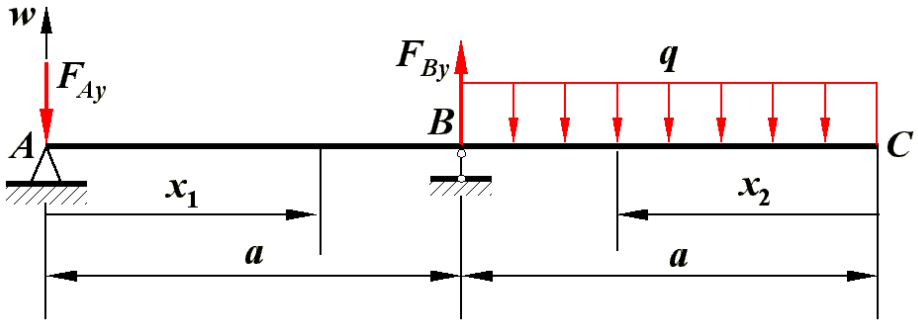


$$x=0,\quad w=0;$$
$$x=l,\quad w=0$$

平面内的静定梁有3个约束，
其中2个与挠度有关，
用于确定两个待定系数。

2、例题

例7-1 图示梁，EI为常数，计算截面C的挠度与转角。



解：1）支反力

$$F_{Ay}=\frac{qa}{2} \qquad F_{By}=\frac{3qa}{2}$$

2）分段建立挠曲轴近似微分方程

$$\frac{\mathrm{d}^2 w_1}{\mathrm{d}x_1^2}=-\frac{qa}{2EI}x_1$$

$$\frac{\mathrm{d}w_1}{\mathrm{d}x_1}=-\frac{qa}{4EI}x_1^2+C_1$$

$$w_1=-\frac{qa}{12EI}x_1^3+C_1x_1+D_1$$

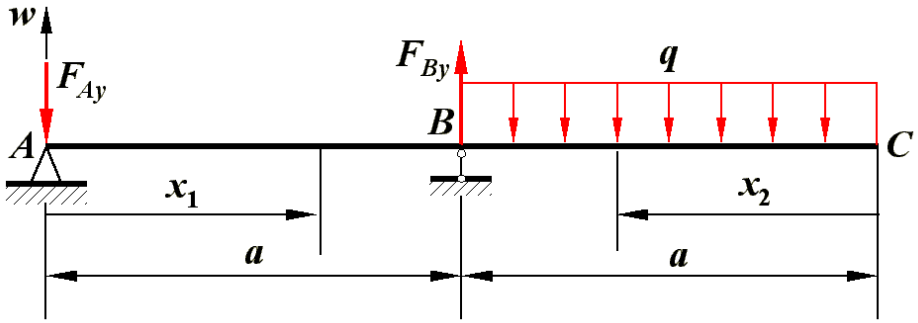
$$\frac{\mathrm{d}^2 w_2}{\mathrm{d}x_2^2}=-\frac{q}{2EI}x_2^2$$

$$\frac{\mathrm{d}w_2}{\mathrm{d}x_2}=-\frac{q}{6EI}x_2^3+C_2$$

$$w_2=-\frac{q}{24EI}x_2^4+C_2x_2+D_2$$

2、例题

例7-1 图示梁，EI为常数，计算截面C的挠度与转角。



解：

$$w_1 = -\frac{qa}{12EI}x_1^3 + C_1x_1 + D_1 \qquad w_2 = -\frac{q}{24EI}x_2^4 + C_2x_2 + D_2$$

3）位移边界条件与连续光滑条件

$$x_1 = 0, \qquad w_1 = 0;$$

$$x_1 = x_2 = a, \quad w_1 = 0;$$

$$w_2 = 0;$$

$$\frac{\mathrm{d}w_1}{\mathrm{d}x_1} = -\frac{\mathrm{d}w_2}{\mathrm{d}x_2}$$

$$\Rightarrow \begin{cases} C_1 = \dfrac{qa^3}{12EI} \\ D_1 = 0 \\ C_2 = \dfrac{qa^3}{3EI} \\ D_2 = -\dfrac{7qa^4}{24EI} \end{cases}$$

$$\theta_C = \theta_2(0) = \frac{qa^3}{3EI} \ (\circlearrowright)$$

$$w_C = w_2(0) = -\frac{7qa^4}{24EI} \ (\downarrow)$$

原理简单，适应面广，过程繁琐

第七章 弯曲变形

7.4 计算梁位移的叠加法

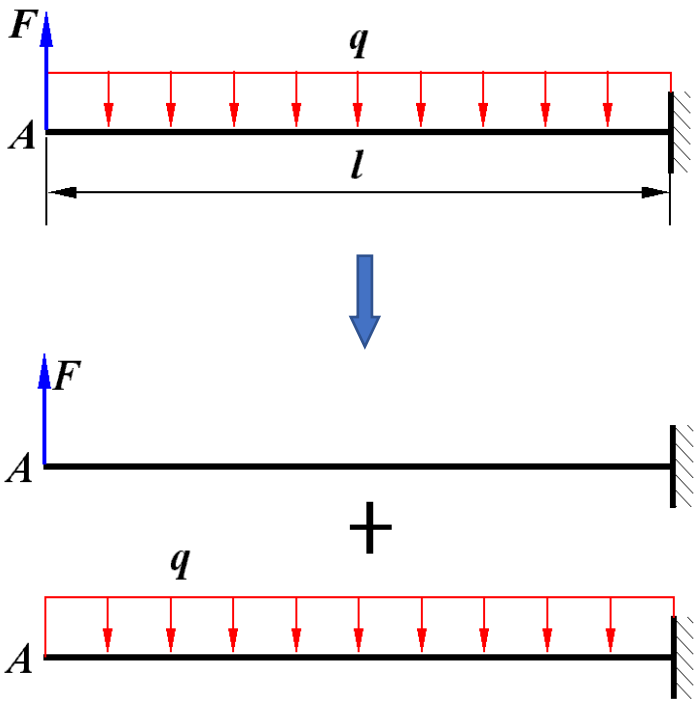
1、载荷叠加法

2、逐段刚化法

3、叠加法应用

1、载荷叠加法

图示梁，EI为常数，求挠度 $w_A = ?$



根据叠加原理

$$w_A = w_{A,F} + w_{A,q}$$

$$= \frac{Fl^3}{3EI} - \frac{ql^4}{8EI} \quad (\uparrow)$$

$$w_{A,F} = \frac{Fl^3}{3EI} \quad (\uparrow)$$

$$w_{A,q} = \frac{ql^4}{8EI} \quad (\downarrow)$$

- 适用范围：小变形、线弹性。
- 六种典型情况：集中力、集中力矩以及均匀分布力作用下的简支梁、悬臂梁（附录E）。

1、载荷叠加法

附录E

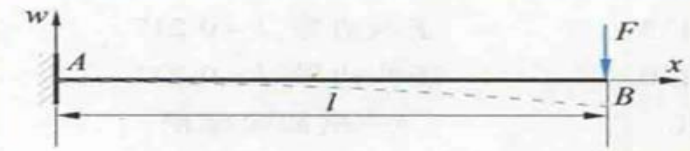
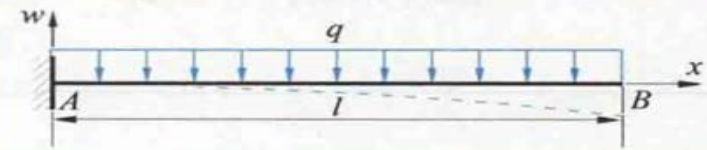
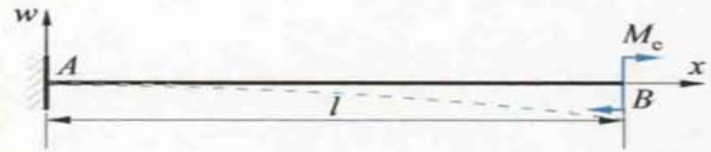
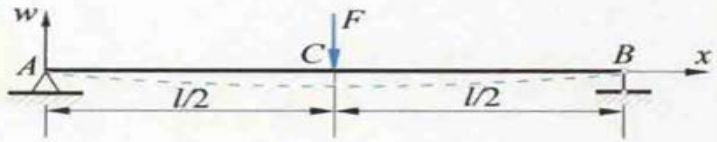
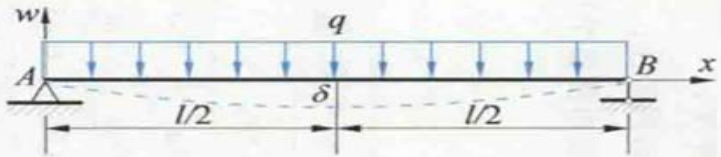
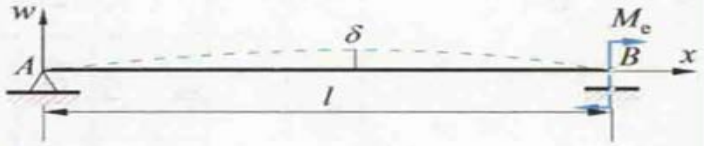
刚度矩阵

$$\begin{Bmatrix} F \\ M_e \end{Bmatrix} = \begin{bmatrix} \dfrac{3EI}{l^3} & \dfrac{2EI}{l^2} \\ \dfrac{2EI}{l^2} & \dfrac{EI}{l^2} \end{bmatrix} \begin{Bmatrix} w_B \\ \theta_B \end{Bmatrix}$$

$$= \begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix} \begin{Bmatrix} w_B \\ \theta_B \end{Bmatrix}$$

k_{ij} 梁的刚度系数

- 引起单位大小的第j个位移，所需要的第i个载荷；
- 与梁的跨度相关。

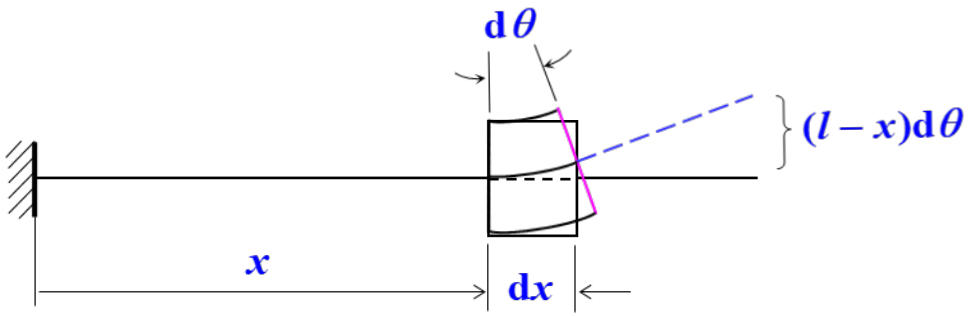
序号	梁的简图	挠度和转角
1		$w_B=-\dfrac{Fl^3}{3EI}$ $\theta_B=-\dfrac{Fl^2}{2EI}$
3		$w_B=-\dfrac{ql^4}{8EI}$ $\theta_B=-\dfrac{ql^3}{6EI}$
4		$w_B=-\dfrac{M_e l^2}{2EI}$ $\theta_B=-\dfrac{M_e l}{EI}$
6		$w_C=-\dfrac{Fl^3}{48EI}$ $\theta_A=-\theta_B=-\dfrac{Fl^2}{16EI}$
8		$\delta=-\dfrac{5ql^4}{384EI}$ $\theta_A=-\theta_B=-\dfrac{ql^3}{24EI}$
9		$\theta_A=\dfrac{M_e l}{6EI}$ $\theta_B=-\dfrac{M_e l}{3EI}$

2、逐段刚化法

——逐段分析求和（逐段软化）

原理：

基于对微段变形与截面位移之关系（影响系数）的认识



- 微段变形 → 截面挠度与转角增加；
- 逐段分析求和 → 截面挠度与转角；
- 逐段刚化梁段，所分析梁段“软化”；
- 与载荷叠加是统一的。

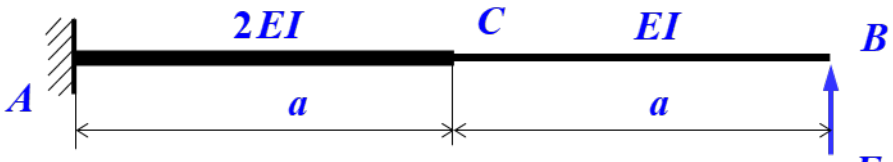
$$\mathrm{d}\theta_B = \mathrm{d}\theta, \quad \mathrm{d}w_B = (l-x)\mathrm{d}\theta$$

$$\mu_{\theta B} = 1, \quad \mu_{wB} = l - x$$

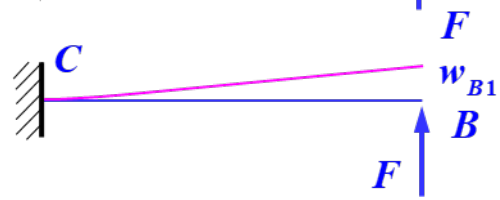
$$\theta_B = \int_0^l \mu_{\theta B}\,\mathrm{d}\theta, \quad w_B = \int_0^l \mu_{wB}\mathrm{d}\theta$$

2、逐段刚化法

图示梁，EI为常数，求挠度 $w_B=?$

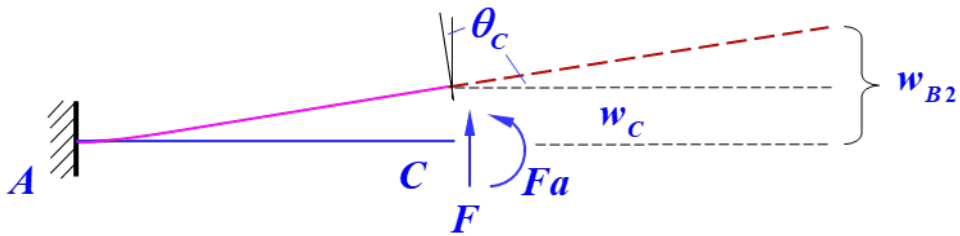


刚化AC段



$$w_{B1}=\frac{Fa^3}{3EI}(\uparrow)$$

刚化CB段

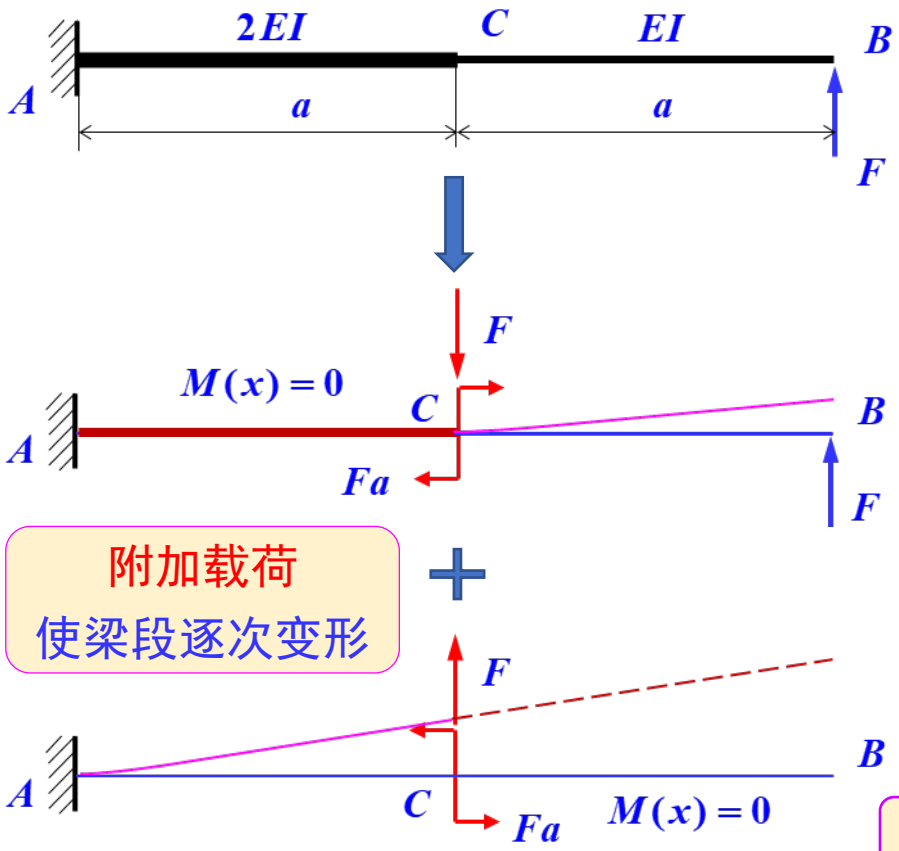


$$w_C=\frac{Fa^3}{3(2EI)}+\frac{Fa\cdot a^2}{2(2EI)}(\uparrow),\quad \theta_C=\frac{Fa^2}{2(2EI)}+\frac{Fa\cdot a}{2EI}(\circlearrowleft)$$

$$w_B=w_{B1}+w_{B2}=w_{B1}+w_C+\theta_C a$$

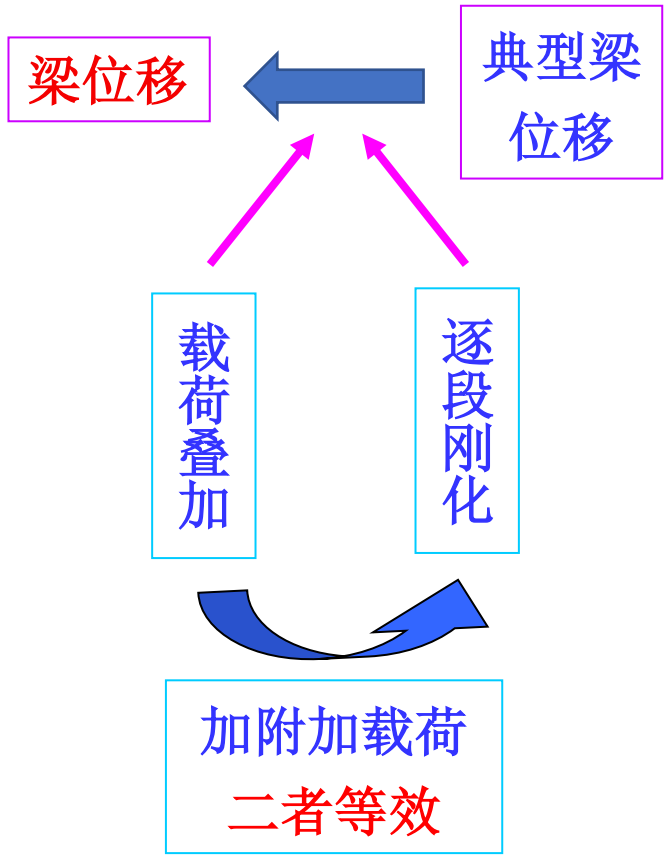
2、逐段刚化法

载荷叠加与逐段刚化等效



附加载荷
使梁段逐次变形

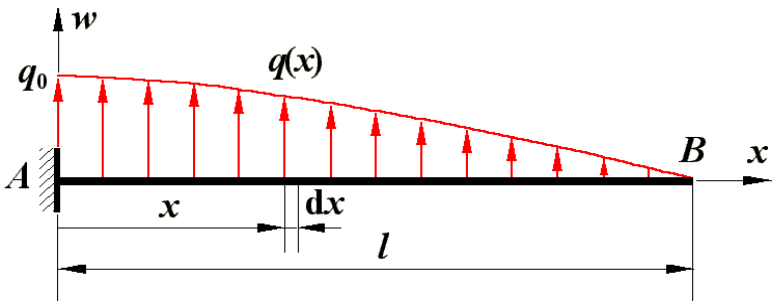
叠加法小结



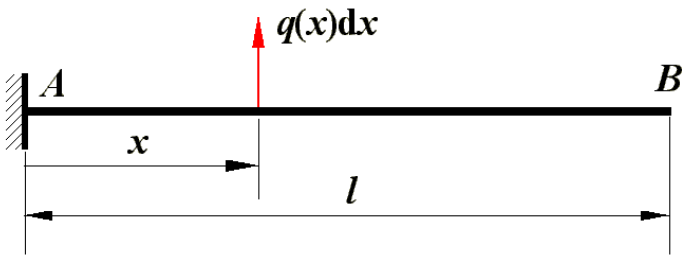
直观、简便、快捷，用于特殊截面

3、叠加法应用

例7-2 图示梁，EI为常数，$q(x)=q_0\cos(\pi x/2l)$，求挠度 w_B=?



解： 取微段dx，受力 $q(x)\mathrm{d}x$



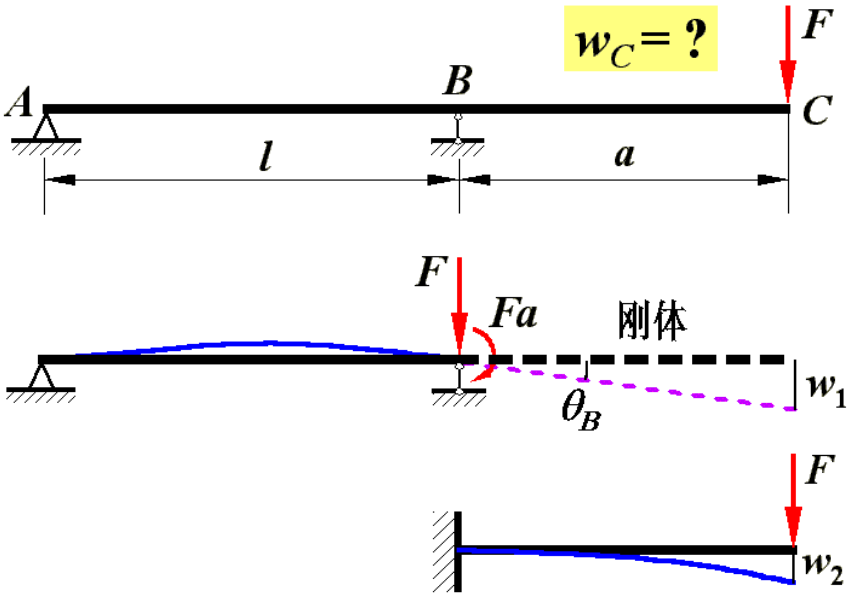
$$\mathrm{d}w_B=\frac{q\mathrm{d}x\cdot x^3}{3EI}+\frac{q\mathrm{d}x\cdot x^2}{2EI}(l-x)$$

$$w_B=\int_0^l \mathrm{d}w_B=\frac{2q_0l^4(\pi^3-24)}{3\pi^4EI}(\uparrow)$$

3、叠加法应用

例7-3 EI为常数，求挠度 w_C=?

解： 刚化BC段



$$\theta_B = \frac{Fa \cdot l}{3EI}, \quad w_1 = \theta_B a = \frac{Fa^2 l}{3EI} (\downarrow)$$

刚化AB段

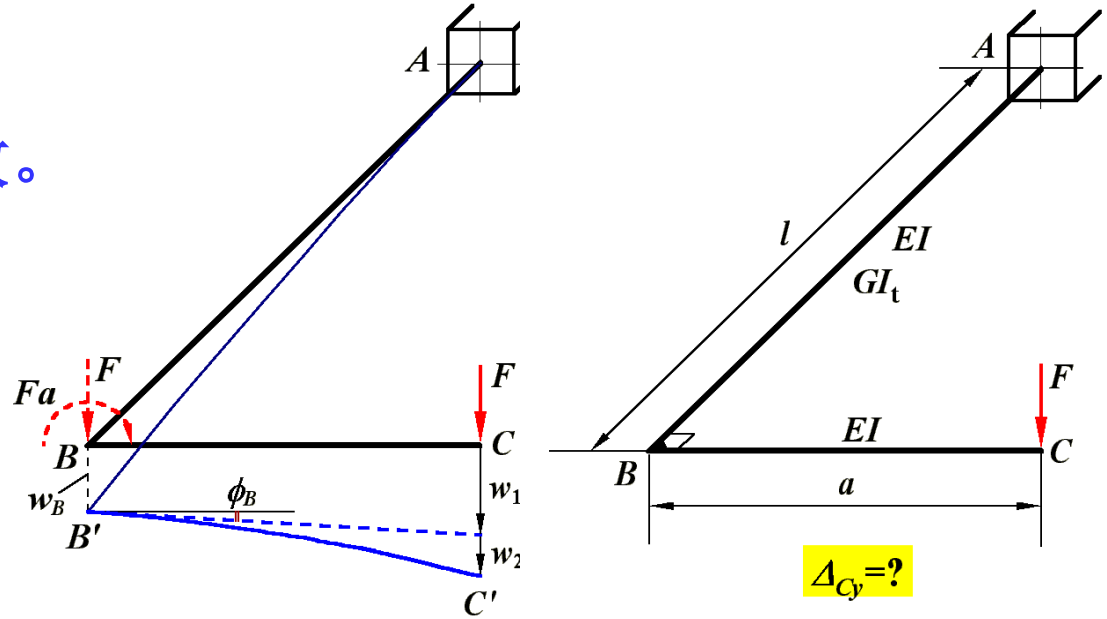
$$w_2 = \frac{Fa^3}{3EI} (\downarrow)$$

$$w_C = w_1 + w_2 = = \frac{Fa^2}{3EI}(l + a)\,(\downarrow)$$

3、叠加法应用

例7-4 图示刚架一端固定， EI、GI_t为常数。求自由端挠度

解：



$$w_1 = w_B + \varphi_B a = \frac{Fl^3}{3EI} + \frac{Fal}{GI_t} \cdot a \ (\downarrow) \qquad w_2 = \frac{Fa^3}{3EI} \ (\downarrow)$$

$$\Delta_{Cy} = w_1 + w_2 = \frac{Fl^3}{3EI} + \frac{Fa^2 l}{GI_t} + \frac{Fa^3}{3EI} \ (\downarrow)$$

3、叠加法应用

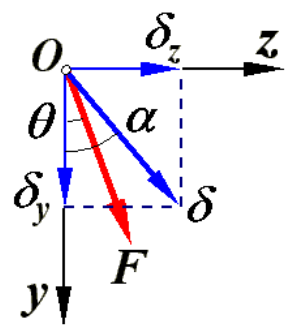
例7-5 求自由端挠度 δ

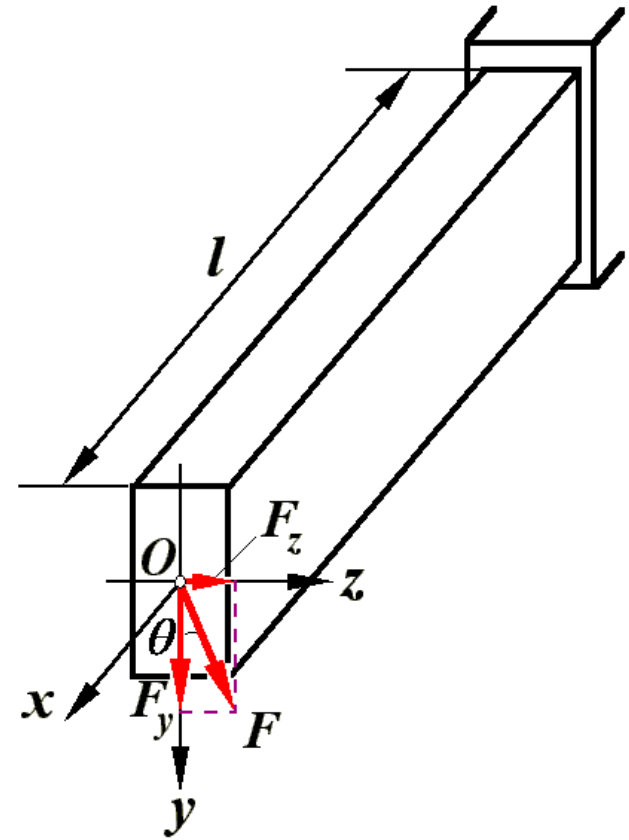
解： $F_y = F\cos\theta \quad F_z = F\sin\theta$

$$\delta_y = \frac{F_y l^3}{3EI_z} = \frac{Fl^3\cos\theta}{3EI_z} \qquad \delta_z = \frac{F_z l^3}{3EI_y} = \frac{Fl^3\sin\theta}{3EI_y}$$

$$\delta = \sqrt{\delta_y^2 + \delta_z^2}$$

$$\tan\alpha = \frac{\delta_z}{\delta_y} = \frac{I_z}{I_y}\tan\theta$$



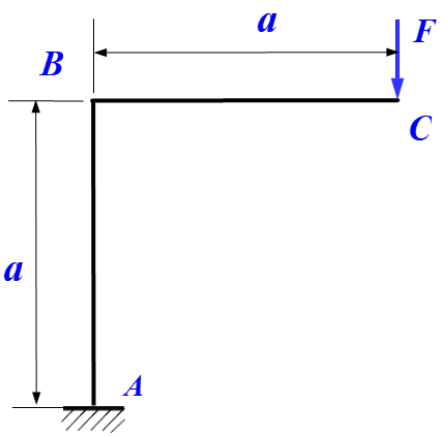


一般情况 $I_y \neq I_z \quad \alpha \neq \theta$

挠曲轴与外力作用面不重合——斜弯曲

3、叠加法应用

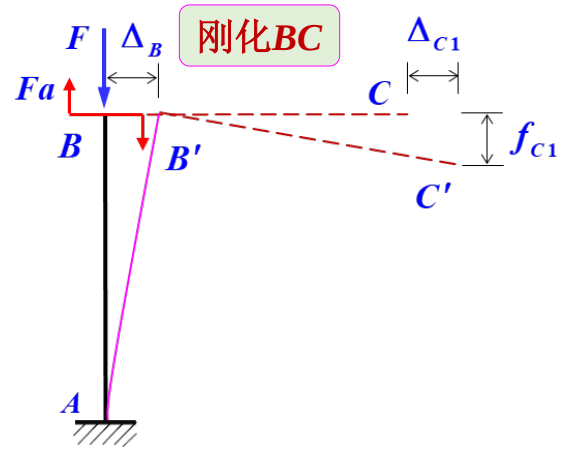
例7-6 EI为常数，求C端位移



$$\Delta l_{AB}=\frac{Fa}{EA},\quad \frac{\Delta l_{AB}}{f_{C1}}=\frac{I}{Aa^2}$$

$$\Delta l_{AB}\ll f_{C1}$$

拉压变形远小于弯曲变形，略去



解：

$$f_{C1}=\theta_B a=\frac{Fa\cdot a}{EI}a$$

$$\Delta_{C1}=\Delta_B=\frac{Fa\cdot a^2}{2EI}$$

$$f_{C2}=\frac{Fa^3}{3EI}$$

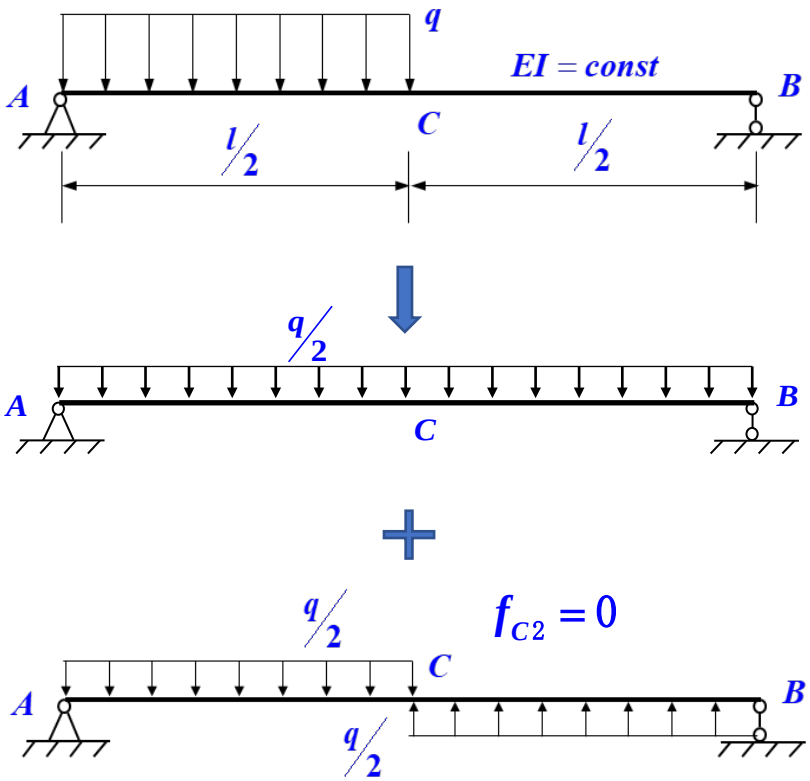
$$f_C=f_{C1}+f_{C2}=\frac{4Fa^3}{3EI}\ (\downarrow),\quad \Delta_C=\Delta_{C1}=\frac{Fa^3}{2EI}\ (\rightarrow)$$

3、叠加法应用

例7-7 求C截面挠度和转角

解：

载荷的对称与反对称分解



$$\theta_{C1}=0,\quad f_{C1}=\frac{5\left(\frac{q}{2}\right)l^4}{384EI}(\downarrow)$$

$$f_{C2}=0$$

$$\theta_{C2}=\frac{\frac{q}{2}\left(\frac{l}{2}\right)^3}{24EI}(\circlearrowleft)$$

$$f_C=f_{C1},\quad \theta_C=\theta_{C2}$$

本讲：弯曲变形

微段变形累加形成梁位移

下讲：简单静不定梁